

Long Division with Remainders

3-Digit $\div$ 1-Digit

❖ What is Long Division with Remainders?

Long division helps us divide large numbers by smaller ones. Sometimes the numbers don't divide evenly — we get **leftovers** called **remainders**. This happens when the divisor doesn't fit perfectly into the dividend!

Key Vocabulary

Term	Meaning	Example
Dividend	The number being divided	456 in $456 \div 7$
Divisor	The number you divide by	7 in $456 \div 7$
Quotient	The result of dividing	65 in $456 \div 7 = 65 \text{ R}1$
Remainder	What's left over	1 in $456 \div 7 = 65 \text{ R}1$

❖ Understanding Remainders

A **remainder** is what's left over when numbers don't divide evenly.

Real-life Example	
Share 10 candies among 3 friends.	$10 \div 3 = ?$
Each friend gets 3 candies ($3 \times 3 = 9$).	Quotient = 3
1 candy is left over — that's the remainder!	Remainder = 1
We write: $10 \div 3 = 3 \text{ R}1$	

❖ Worked Examples — Step by Step

The 4-Step Method: Divide → Multiply → Subtract → Bring Down

Step	Action	What to do
1	DIVIDE	How many times does the divisor go into the current number?
2	MULTIPLY	Multiply the divisor by the digit you just wrote.
3	SUBTRACT	Subtract the product from the current number.
4	BRING DOWN	Bring down the next digit and repeat!

Example 1: $456 \div 7 = 65 \text{ R}1$

- 1 **DIVIDE:** $4 \div 7 = 0$, so look at 45. $45 \div 7 = 6$ (write 6 in quotient)
- 2 **MULTIPLY:** $6 \times 7 = 42$
- 3 **SUBTRACT:** $45 - 42 = 3$
- 4 **BRING DOWN** the 6 → new number is 36
- 5 **DIVIDE:** $36 \div 7 = 5$ (write 5 in quotient)
- 6 **MULTIPLY:** $5 \times 7 = 35$
- 7 **SUBTRACT:** $36 - 35 = 1$ → Remainder = 1 ✓ Answer: 65 R1

Example 2: $789 \div 4 = 197 \text{ R}1$

- 1 DIVIDE: $7 \div 4 = 1$ (write 1)
- 2 MULTIPLY & SUBTRACT: $1 \times 4 = 4$. $7 - 4 = 3$
- 3 BRING DOWN 8 $\rightarrow$ 38. $38 \div 4 = 9$ (write 9)
- 4 MULTIPLY & SUBTRACT: $9 \times 4 = 36$. $38 - 36 = 2$
- 5 BRING DOWN 9 $\rightarrow$ 29. $29 \div 4 = 7$ (write 7)
- 6 MULTIPLY & SUBTRACT: $7 \times 4 = 28$. $29 - 28 = 1 \rightarrow$ Remainder = 1
- 7 Answer: $789 \div 4 = 197 \text{ R}1$ ✓

Example 3: $523 \div 6 = 87 \text{ R}1$

- 1 DIVIDE: $5 \div 6 = 0$, so look at 52. $52 \div 6 = 8$ (write 8)
- 2 MULTIPLY & SUBTRACT: $8 \times 6 = 48$. $52 - 48 = 4$
- 3 BRING DOWN 3 $\rightarrow$ 43. $43 \div 6 = 7$ (write 7)
- 4 MULTIPLY & SUBTRACT: $7 \times 6 = 42$. $43 - 42 = 1 \rightarrow$ Remainder = 1
- 5 Answer: $523 \div 6 = 87 \text{ R}1$ ✓

❖ Tips and Tricks for Long Division

Tip	Details
❖ Break it down	Work one digit at a time — don't rush the whole problem at once.
❖ Check multiply	Quotient digit $\times$ divisor must be $\leq$ the current working number.
❖ Remainder rule	The remainder must always be smaller than the divisor.
❖ Estimate first	Round numbers to make a quick guess before working through the steps.
❖ Verify your answer	Quotient $\times$ divisor + remainder = original dividend.

❖ Real-World Uses of Division with Remainders

Situation	How Remainders Help
-----------	---------------------

Sharing sweets among friends	Remainders show how many are left after equal sharing.
Packing items into boxes	Tells you how many items won't fill a complete box.
Splitting money equally	Leftover coins or notes can be saved or given as change.
Arranging seats in rows	Leftover seats after all complete rows are filled.

⇒ Practice Problems — Try It Yourself!

Solve each long division problem. Show all your working steps in the space provided!

Problem 1

$$348 \div 5 = ?$$

Working Space:

Answer: _____

Problem 2

$$672 \div 8 = ?$$

Working Space:

Answer: _____

Problem 3

$$519 \div 4 = ?$$

Working Space:

Answer: _____

Problem 4

$$836 \div 9 = ?$$

Working Space:

Answer: _____

Problem 5

$$247 \div 3 = ?$$

Working Space:

Answer: _____

Problem 6

$$965 \div 7 = ?$$

Working Space:

Answer: _____

Answer Key (for teachers/parents): $348 \div 5 = 69 \text{ R}3$ · $672 \div 8 = 84 \text{ R}0$ · $519 \div 4 = 129 \text{ R}3$ · $836 \div 9 = 92 \text{ R}8$ · $247 \div 3 = 82 \text{ R}1$ · $965 \div 7 = 137 \text{ R}6$